

# How does this even work?

## Queens example

### Solution/1.

Base: empty list  $[]$

Recursive: on list  $[[X, Y] | R]$

True if

- 1.)  $\text{solution}(R)$
- 2.)  $\text{member}(Y, [1, 2, 3, 4, 5])$
- 3.)  $\text{noattack}(\underline{[X, Y]}, \underline{R})$

$\text{solution}(R)$  checks that the rest of the queens' placement is valid.

Then

$\text{member}(Y, [1, 2, 3, 4, 5])$  assigns the queen a row value

$\text{noattack}$  ensures this placed queen doesn't attack any other queen.

$\text{noattack}$  checks whether the head of the **remaining** list conflicts with **this** queen.

Then it recurses to check if the rest of queens after the head are valid.

Now for the knight example...

